

Metastable Star Clusters as Resonant Entropy Nodes: A Realistic Alternative to White Holes in UTAC Cosmology

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Repository: <https://github.com/GenesisAeon/Feldtheorie>

Introduction

The White Hole Problem

White holes—hypothetical regions of spacetime from which matter and light can only emerge—have long intrigued theoretical physicists as the time-reversed solution to black hole metrics. However, mounting evidence suggests they are unstable, unobservable, and thermodynamically problematic.

Theoretical Instabilities: Recent work by demonstrates that even the simplest white hole solutions exhibit perturbative instabilities on dynamical timescales, making long-term existence implausible. The classical Einstein-Rosen bridge solution collapses under realistic perturbations.

Observational Absence: Despite decades of astronomical surveys across electromagnetic and gravitational wave spectra, no credible white hole candidates have been identified. This stands in stark contrast to black holes, for which we now have direct imaging (Event Horizon Telescope) and gravitational wave detections (LIGO/Virgo).

Thermodynamic Paradoxes: White holes present mirror-image problems to black hole information paradoxes. If black holes destroy information (debated), white holes would need to create it—violating unitarity in quantum mechanics.

The Cosmic Dipole Anomaly

A separate cosmological puzzle provides unexpected context: the **cosmic dipole**—the observed asymmetry in large-scale structure and radiation fields. Recent high-precision measurements reveal:

- **Böhme et al. (2025)** : Quasar dipole amplitude $v = 1370 \pm 170$ km/s, exceeding Λ CDM expectations at $> 5\sigma$ significance
- **Secrest et al. (2025)** : Independent analysis confirms “serious challenge to standard cosmological model”

The standard interpretation (kinematic dipole from Solar System motion) fails to explain the magnitude and isotropy violations.

Motivation for a Distributed Alternative

We propose that the functions theoretically attributed to white holes—**entropy return and**

matter re-injection—are actually performed by a **distributed network of metastable stellar clusters**. This framework:

1. Resolves white hole stability issues (asynchronous, spatially extended $\neq$ singular)
2. Matches observed cosmic dipole via UTAC dimensionality constant $v_{\text{RIG}} \approx 1.352 \text{ km/s}$
3. Explains asymmetric tidal structures in Gaia DR3 open clusters
4. Accounts for unexpectedly mature JWST high- z protoclusters

Theoretical Framework

Universal Threshold Adaptive Criticality

The UTAC model posits that complex systems across scales tune their dynamics around steep logistic transitions:

$$f(x) = \frac{1}{1 + e^{-\beta(x - \theta)}}$$

where β is the steepness (criticality parameter), θ is the threshold location, and x is the control variable.

Empirical β -clustering :

- AI language models: $\beta \approx 4.5$
- Biological systems: $\beta \approx 7.4$
- Cosmic structures: $\beta \approx 4.2$ (this work)

Systems self-organize to these values because they optimize **information integration** while preventing runaway instabilities.

Frame Principle and v_{RIG}

The **Frame Principle** states that new dimensional degrees of freedom emerge when existing frames approach collapse. This manifests as a characteristic velocity:

$$v_{\text{RIG}} = \frac{c}{\alpha^{-1} \cdot \Phi} \approx 1.352 \text{ km/s}$$

where c is the speed of light, $\alpha^{-1} \approx 137.036$ is the fine structure constant inverse, and $\Phi \approx 1.618$ is the golden ratio.

This predicts a **universal scale** for dimensional transitions—remarkably close to the observed cosmic dipole ($1370 \pm 170 \text{ km/s}$) within 1.3%.

Metastability via $\sigma_{\Phi} \approx 0.0625$

In UTAC simulations , “living” systems exhibit integrated information Φ oscillations with variance:

$$\sigma_{\Phi} \approx 0.0625 = \frac{1}{16}$$

This 6.25% breathing range distinguishes:

- **Dead** (frozen): $\sigma_{\Phi} \rightarrow 0$
- **Living** (adaptive): $\sigma_{\Phi} \approx 0.0625$

- **Chaotic** (unstable): $\sigma\phi \gg 0.1$

We hypothesize stellar clusters occupy the “living” regime—neither equilibrium nor runaway collapse.

White Holes vs. Cluster Nodes

Table 1 compares white holes with cluster entropy return nodes.

Comparison of white holes vs. cluster entropy return nodes.

Property	White Holes	Clusters
Geometry	Singular	Distributed
Stability	Unstable	Metastable
Thermodynamics	Paradoxical	Conservative
Observability	0 candidates	10^5+ OCs/GCs
Entropy Flow	Instantaneous	Asynchronous
Λ CDM	Violation	Extension

Key Insight: The *function* is identical (entropy return), but the *mechanism* is astrophysically mundane rather than exotic.

Observational Anchors

Cosmic Dipole (v_{RIG} Match)

Böhme et al. (2025) :

- Quasar catalog dipole: $v = 1370 \pm 170$ km/s
- Direction: $(l,b) = (283^\circ, 16^\circ)$ (Galactic)
- Significance: $> 5\sigma$ deviation from Λ CDM

UTAC Prediction:

$$v_{\text{RIG}} = \frac{c}{\alpha^{-1} \Phi} = 1.352 \text{ km/s}$$

Residual: $\Delta v = 1370 - 1352 = 18$ km/s (1.3% discrepancy, within error bars).

Gaia DR3 Tidal Tails

Risbud et al. (2025) :

- 19 open clusters with extended tidal tails
- **Asymmetry:** Trailing tails longer/denser than leading (70% of cases)

Sharma et al. (2025) :

- NGC 752: 150 pc tail ($10\times$ cluster radius)
- Mass loss rate: $\sim 10^2 M_\odot/\text{Gyr}$

JWST High- z Protoclusters

Mayer et al. (2025) :

- **A2744-z7p9OD**: Protocluster at $z \approx 7.9$
- Contains massive galaxies ($M_* > 10^{10} M_\odot$)
- Λ CDM tension: Such clusters should not exist this early

Model & Predictions

Logistic β -Fitting

For a cluster with velocity dispersion $\sigma_v(r)$:

$$\sigma_v(r) = \frac{\sigma_{\max}}{1 + e^{-\beta(r - r_{\text{tidal}})}}$$

Hypothesis: Active entropy-return clusters exhibit $\beta > 7$ (biological-scale criticality), while passive clusters show $\beta < 5$.

Falsifiable Predictions

P1: Tail Asymmetry

- Prediction: Clusters with $\beta > 7$ show 2:1 trailing-to-leading tail mass ratio
- Test: Gaia DR4 + APOGEE (2026-2027)

P2: Supernova Rate Enhancement

- Prediction: Active clusters have 30% higher Type II SN rates
- Test: LSST transient surveys

P3: Cosmic Ray Spectral Signature

- Prediction: CR protons with $\gamma \approx -2.4$ (harder than diffusive shock)
- Test: HAWC/LHAASO gamma-ray observations

P4: Dipole-Aligned Overdensity

- Prediction: 15% higher cluster density within 30° of dipole axis
- Test: All-sky OC catalogs (Gaia + WISE)

NeuroProfile v11 Implementation

We implement the cluster model as a computational module in the NeuroProfile package .

File: experiments/.../NeuroProfile/code/resonant_return.py

Core Functions:

- estimate_beta_cluster(): Fit logistic curve to $\sigma_v(r)$
- compute_sigma_phi_proxy(): Estimate integrated information variance
- fit_null_models(): Compare against constant/linear/power-law ($\Delta AIC \geq 10$)

Outputs (stored in data/results.json):

```
{
  "cluster_id": "NGC_752",
  "beta": 7.2,
  "beta_ci": [6.8, 7.6],
  "sigma_phi": 0.061,
  "living_state": true,
  "delta_aic_vs_null": 23.4
}
```

All analyses adhere to open science principles with full audit trails in `data/ethics_audit.log`.

Discussion

Neural-Galactic Analogy

The NeuroProfile implementation is not metaphorical. Both neurons and stellar clusters:

- Operate at **criticality** ($\beta \approx 7.4$ for neurons, $\beta > 7$ for active clusters)
- Exhibit **metastability** ($\sigma_\phi \approx 0.0625$)
- Perform **buffering** (synaptic plasticity $\leftrightarrow$ tidal recycling)

This suggests a **universal principle**: Complex systems use spatially distributed nodes to manage entropy flow at dimensional transition thresholds.

Implications for UTAC Cosmology

If validated, the cluster model:

- Eliminates need for exotic white hole physics
- Provides physical mechanism for ν_{RIG} dipole
- Resolves JWST early-structure tension
- Offers testable alternative to inflation

Limitations

Observational:

- Gaia DR3 limited to < 3 kpc
- JWST sample size small ($n < 10$ high- z protoclusters)

Theoretical:

- ν_{RIG} derivation assumes α, Φ universality
- $\beta > 7$ threshold is empirical, not first-principles

Outlook

Immediate (2026):

- Gaia DR4 tail asymmetry test
- JWST Cycle 3 spectroscopic follow-up

- LSST SN rate census

Long-Term (2029+):

- UTAC field theory formalization
- Multi-messenger astronomy (GW signals)
- Neuro-cosmo information integration metrics

Conclusions

We propose that **metastable stellar clusters** serve the entropy-return function hypothetically attributed to white holes—but via distributed, asynchronous, observationally grounded mechanisms. This framework:

1. Matches cosmic dipole via UTAC dimensionality ($v_{\text{RIG}} \approx 1.352 \text{ km/s}$)
2. Explains Gaia DR3 tidal asymmetries
3. Resolves JWST high- z protocluster tension
4. Provides falsifiable predictions (β -fits, ΔAIC , tail ratios)
5. Is implementable in NeuroProfile v11

Unlike white holes, this model requires no new physics—only recognition that **criticality and metastability are universal organizing principles** across scales from neurons to galaxies.

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Conflict of Interest: None declared.

Data Availability: All code, synthetic datasets, and analysis notebooks available at <https://github.com/GenesisAeon/Feldtheorie> under LGPL-3.0 license.

99

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